Cohomological dimension of products in comodules over $\mathcal{O}(\mathbb{G}_a)$

Let

$$k = \mathbb{F}_p, \qquad A = \mathcal{O}(\mathbb{G}_a) = k[t].$$

Let

$$\mathcal{A} = A\text{-comod}, \qquad \mathcal{D} = D(\mathcal{A}), \qquad \widehat{\mathcal{D}} = \widehat{D}(\mathcal{A}).$$

Let $k \in \mathcal{A}$ denote the trivial comodule.

The goal is to show that countable products have infinite cohomological dimension in both the separated and left-completed derived categories.

1 The small cofree resolution

The distribution algebra of $\mathbb{G}_a$ is

$$\mathrm{Dist}(\mathbb{G}_a) \cong k[u_0, u_1, u_2, \dots]/(u_0^p, u_1^p, u_2^p, \dots).$$

Here u_i is dual to the primitive monomial t^{p^i}. Its action on $A = k[t]$ is the Hasse derivative

$$u_i \cdot t^N = \binom{N}{p^i} t^{N-p^i},$$

with the convention that the result is 0 if $N < p^i$.

For a finite-support sequence

$$\mathbf{r} = (r_0, r_1, r_2, \dots), \qquad r_i \geq 0,$$

write

$$|\mathbf{r}| = \sum_i r_i.$$

Let V_n be the k-vector space with basis

$$e_{\mathbf{r}}, \qquad |\mathbf{r}| = n.$$

Set

$$I^n = A \otimes V_n.$$

This is a cofree A-comodule.

Let δ_i be the sequence with 1 in the i-th position and 0 elsewhere. Define

$$\epsilon_i(\mathbf{r}) = r_0 + \cdots + r_{i-1}.$$

For each $\mathbf{r}$, put

$$\theta_i(\mathbf{r}) = \begin{cases} u_i, & r_i \text{ even}, \\ u_i^{p-1}, & r_i \text{ odd}. \end{cases}$$

Define

$$d : I^n \to I^{n+1}$$

by

$$d(a \otimes e_{\mathbf{r}}) = \sum_{i \geq 0} (-1)^{\epsilon_i(\mathbf{r})} \big(\theta_i(\mathbf{r}) \cdot a\big) \otimes e_{\mathbf{r}+\delta_i}.$$

The sum is finite, since $a \in k[t]$ has finite degree.

Lemma 1. *Together with the map*

$$k \to I^0 = A \otimes V_0, \qquad 1 \mapsto 1 \otimes e_{\mathbf{0}},$$

the complex

$$0 \to k \to I^0 \to I^1 \to I^2 \to \cdots$$

is a cofree resolution of the trivial comodule k.

Proof. First, $d^2 = 0$. In one variable, the two consecutive maps are multiplication by u_i and u_i^{p-1}, so their composite is multiplication by

$$u_i^p = 0.$$

For two different variables u_i, u_j, the operators commute, and the Koszul signs in the formula above make the two possible orders cancel.

It remains to note exactness. For $N \geq 1$, let

$$A_{\langle N \rangle} = \mathrm{span}_k\{1, t, \ldots, t^{p^N - 1}\} \subset A.$$

This is the finite subcoalgebra dual to

$$k[u_0, \ldots, u_{N-1}]/(u_0^p, \ldots, u_{N-1}^p).$$

The part of the displayed complex using only the variables $u_0, \ldots, u_{N-1}$ is dual to the tensor product of the standard two-periodic resolutions

$$\cdots \to k[u_i]/u_i^p \xrightarrow{u_i^{p-1}} k[u_i]/u_i^p \xrightarrow{u_i} k[u_i]/u_i^p \to k.$$

Hence the finite-stage complex is exact.

The full displayed complex is the filtered union of these finite-stage complexes, and filtered colimits of k-vector spaces are exact. Therefore the full complex is exact. □

We will use one simple consequence of the formula for d.

For a finite subset $S \subset \mathbb{N}$, write

$$e_S := e_{\mathbf{1}_S} \in V_{|S|}.$$

Thus e_S is the squarefree basis vector corresponding to S.

Lemma 2 (Squarefree coefficient lemma). *Let $S \subset \mathbb{N}$ be finite with $|S| = q$. Let $W \subset A$ be a finite-dimensional subspace such that*

$$u_i \cdot W = 0 \qquad \text{for every } i \in S.$$

Then, for every element

$$z \in W \otimes V_{q-1} \subset A \otimes V_{q-1},$$

the coefficient of e_S in $dz \in A \otimes V_q$ is zero.

Proof. A term can contribute to the squarefree basis vector e_S only by starting from

$$e_{S\setminus\{i\}}$$

for some $i \in S$, and then increasing the i-th coordinate from 0 to 1. In that case the operator applied to the outer A-coefficient is u_i, up to sign. By assumption u_i kills W, so every such contribution is zero. □

We also record the following elementary finiteness fact.

Lemma 3. *If $W \subset A = k[t]$ is finite-dimensional, then*

$$u_i \cdot W = 0$$

for all sufficiently large i.

Proof. Choose N such that W is contained in the span of

$$1, t, \ldots, t^N.$$

If $p^i > N$, then

$$u_i \cdot t^a = \binom{a}{p^i} t^{a-p^i} = 0$$

for every $0 \le a \le N$. Hence $u_i \cdot W = 0$. □

2 The separated derived category

Use cohomological shifts with

$$(I^\bullet[i])^n = I^{n+i}.$$

Consider

$$P_{\text{sep}} := \prod_{i\ge 0}^{\mathcal{D}} k[i].$$

Replacing each factor $k[i]$ by the shifted cofree resolution $I^\bullet[i]$, the degree n term is

$$C^n_{\text{sep}} = A \otimes \prod_{m \ge \max\{n,0\}} V_m.$$

The differential is induced by the maps

$$A \otimes V_m \to A \otimes V_{m+1}$$

from the small resolution.

Proposition 1. *For every $q \geq 1$,*

$$H^q(P_{\mathrm{sep}}) \neq 0.$$

Proof. Fix $q \geq 1$. Define

$$x_q = 1 \otimes \left(e_{\{m,m+1,\ldots,2m-1\}}\right)_{m \geq q} \in A \otimes \prod_{m \geq q} V_m = C^q_{\mathrm{sep}}.$$

Since every positive distribution kills $1 \in A$, we have

$$dx_q = 0.$$

We show that x_q is not a boundary. Suppose

$$x_q = dy$$

for some

$$y \in C^{q-1}_{\mathrm{sep}} = A \otimes \prod_{m \geq q-1} V_m.$$

Using the cofree product description, write

$$y = \sum_{\ell=1}^{N} a_\ell \otimes (v_{\ell,m})_{m \geq q-1}.$$

Let

$$W = \mathrm{span}_k\{a_1, \ldots, a_N\} \subset A.$$

Choose $m \geq q$ so large that

$$u_i \cdot W = 0 \qquad \text{for every } i \in \{m, m+1, \ldots, 2m-1\}.$$

Now project the equality $x_q = dy$ to the internal-degree m component

$$A \otimes V_m.$$

The coefficient of

$$e_{\{m,m+1,\ldots,2m-1\}}$$

on the left side is 1. On the right side, the squarefree coefficient lemma says that this coefficient is 0, because all relevant u_i's kill W. This is a contradiction.

Therefore x_q is not a boundary, and

$$H^q(P_{\mathrm{sep}}) \neq 0.$$

$\square$

Remark 1. *This is the small-resolution form of the calculation appearing in Neeman's example for the separated derived category of representations of $\mathbb{G}_a$ over $\mathbb{F}_p$. The class above is the same obstruction as in the terminal-coface/cobar calculation, but the small resolution makes the non-boundary argument visible as a squarefree coefficient calculation.*

3 The left-completed derived category

Now consider the product of countably many copies of the heart object k:

$$P_{\mathrm{lc}} := \prod_{s\geq 1}^{\widehat{\mathcal{D}}} k.$$

Using the cofree resolution $I^\bullet$ in each factor gives the complex

$$C^n_{\mathrm{lc}} = A \otimes \prod_{s\geq 1} V_n.$$

The differential applies the small-resolution differential in each component.

Proposition 2. *For every $q \geq 0$,*

$$H^q(P_{\mathrm{lc}}) \neq 0.$$

In particular, in the left-completed derived category, a countable product of objects concentrated in degree 0 need not be cohomologically bounded above.

Proof. For $q = 0$, the product of the copies of k in the heart is nonzero, so

$$H^0(P_{\mathrm{lc}}) \neq 0.$$

Now fix $q \geq 1$. Define

$$x_q = 1 \otimes \left(e_{\{sq,sq+1,\ldots,sq+q-1\}}\right)_{s\geq 1} \in A \otimes \prod_{s\geq 1} V_q = C^q_{\mathrm{lc}}.$$

Again, every positive distribution kills 1, so

$$dx_q = 0.$$

Suppose, for contradiction, that

$$x_q = dy$$

for some

$$y \in C^{q-1}_{\mathrm{lc}} = A \otimes \prod_{s\geq 1} V_{q-1}.$$

Write

$$y = \sum_{\ell=1}^{N} a_\ell \otimes (v_{\ell,s})_{s\geq 1}.$$

Let

$$W = \mathrm{span}_k\{a_1, \ldots, a_N\} \subset A.$$

Choose s so large that

$$u_i \cdot W = 0 \qquad \text{for every } i \in \{sq, sq+1, \ldots, sq+q-1\}.$$

Project the equality $x_q = dy$ to the s-th component

$$A \otimes V_q.$$

The coefficient of

$$e_{\{sq,sq+1,\dots,sq+q-1\}}$$

on the left side is 1. On the right side, the squarefree coefficient lemma says that this coefficient is 0, because all the relevant u_i's kill W. This contradiction proves that x_q is not a boundary.

Therefore

$$H^q(P_{\text{lc}}) \neq 0.$$

Since $q \geq 1$ was arbitrary, P_{lc} has cohomology in arbitrarily high degrees. $\square$